Recommender Systems
lecture 6: Neural recommenders

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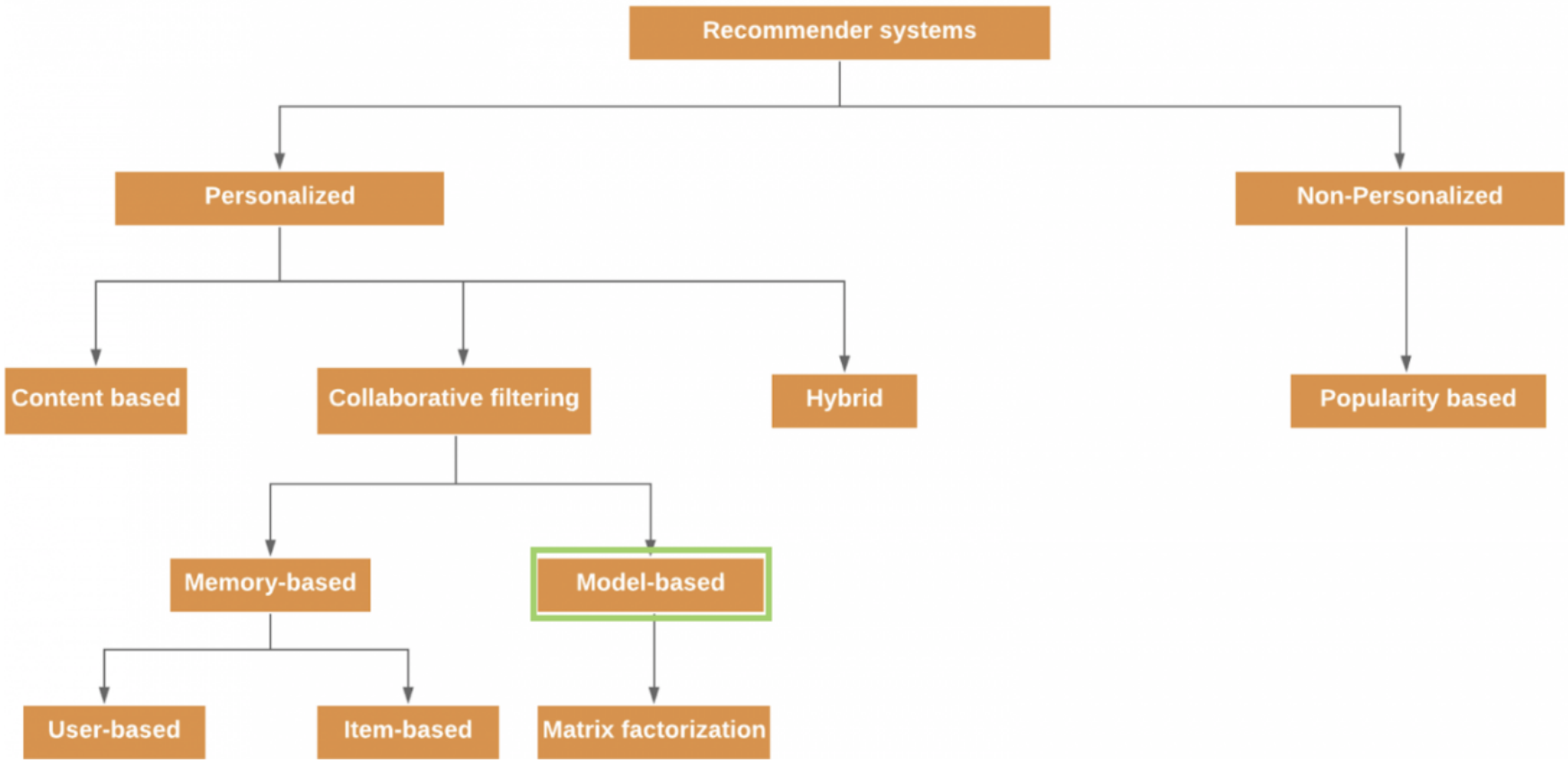
Recap: taxonomy



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Plan for today

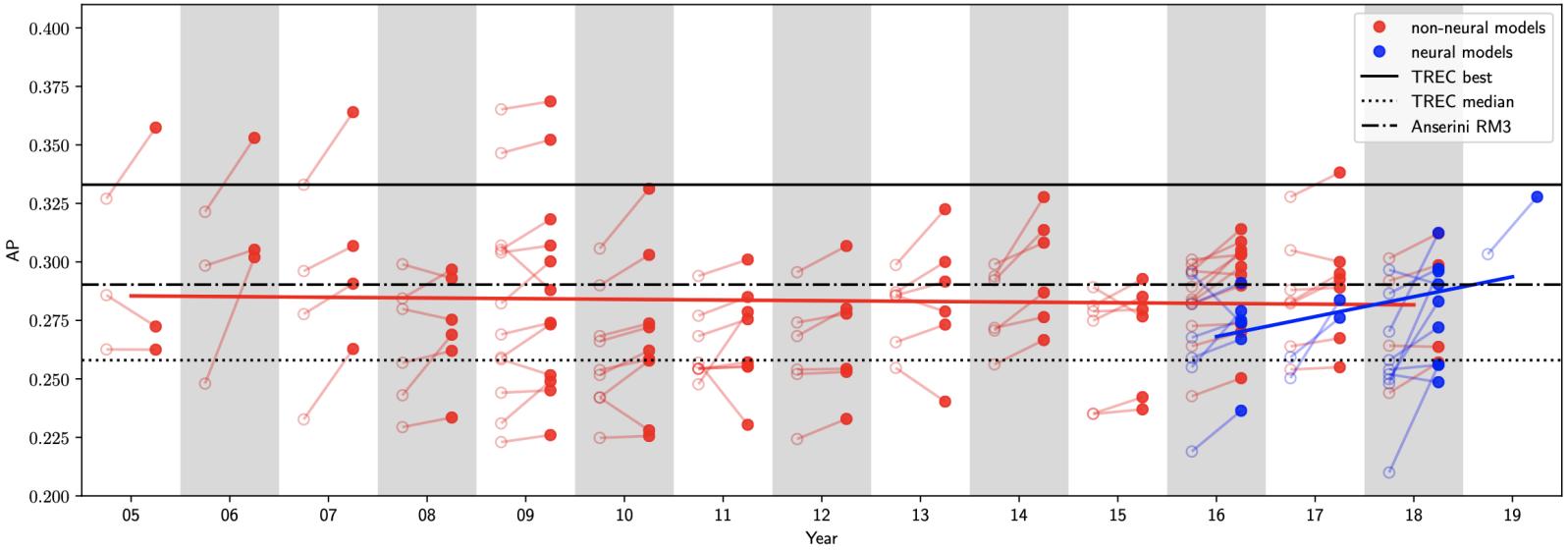
1. NeuroMF
2. MultVAE
3. Sequential models

«Neural hype», 2019

M. Dacrema et al. Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches (RecSys '19) - best paper

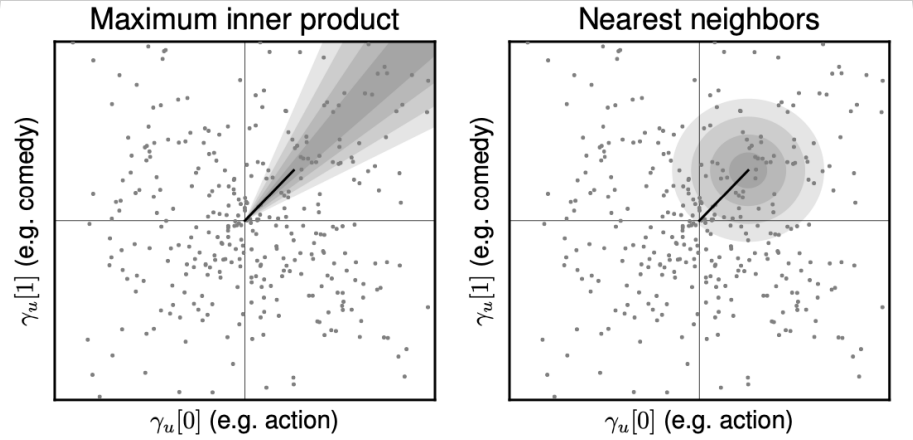
- considered 18 algorithms from top conferences in the last years
- only 7 of them could be reproduced with reasonable effort
- 6 can often be outperformed with comparably simple heuristic methods (nearest-neighbor or graph-based techniques)

W. Yang et al. Critically Examining the "Neural Hype": Weak Baselines and the Additivity of Effectiveness Gains from Neural Ranking Models. (SIGIR'19)



Neural Collaborative Filtering (NCF)

Motivation: generalization of MF (dot product)



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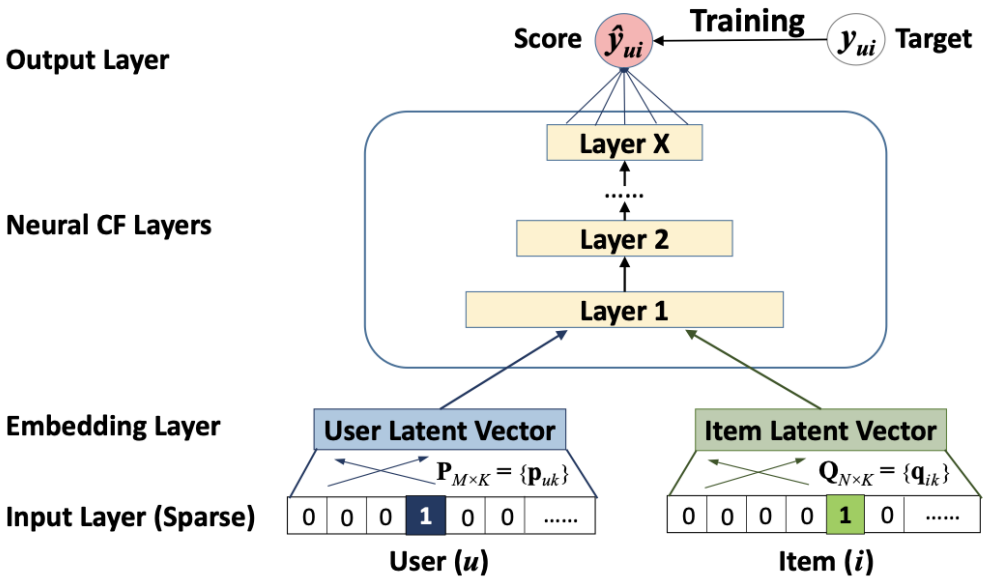
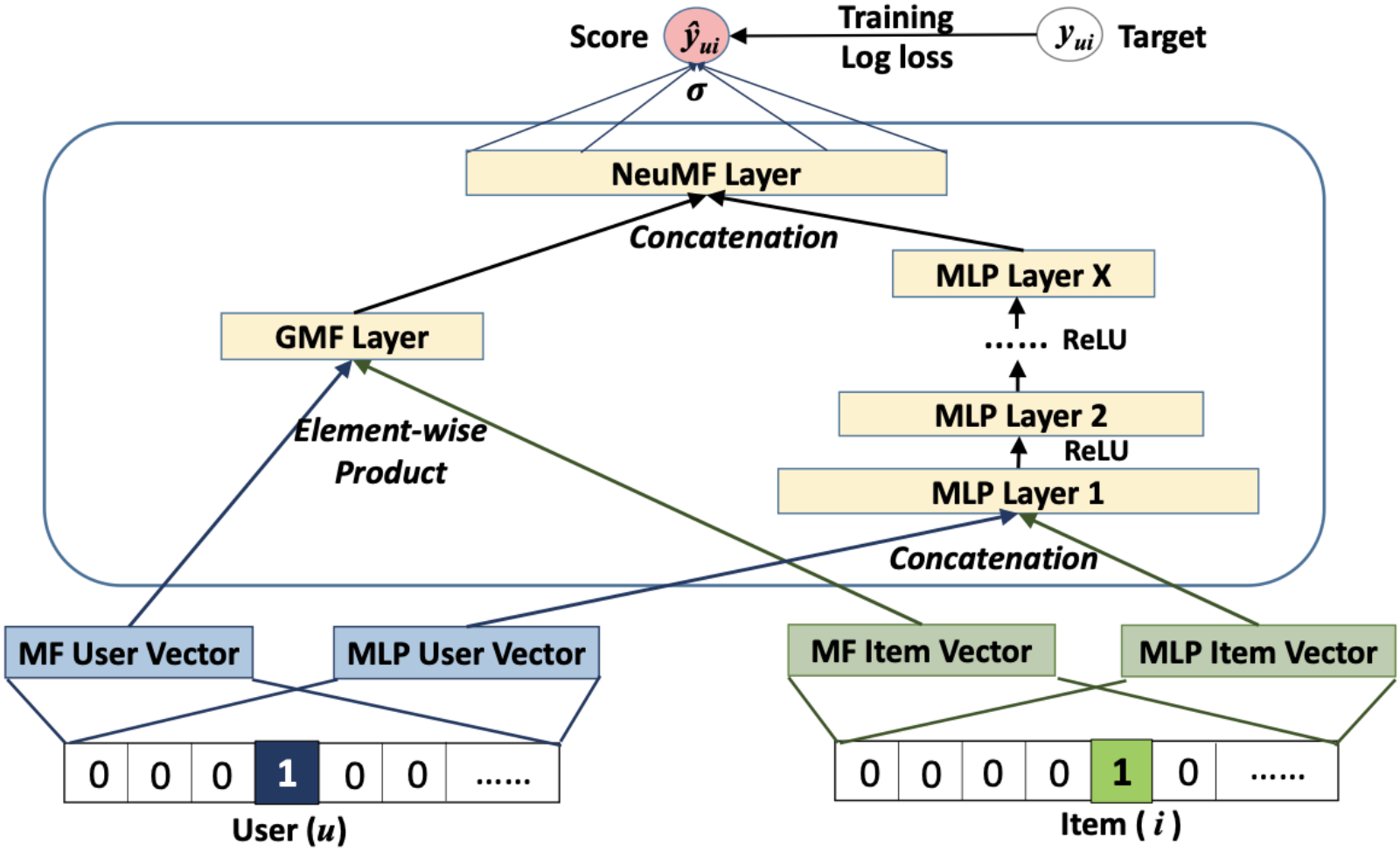
Architecture:



NeuroMF



X. He et al. (WWW 2017). Neural Collaborative Filtering

NeuroMF vs Matrix Factorization

Method	Movielens		Pinterest		Result
	HR@10	NDCG@10	HR@10	NDCG@10	from
Popularity	0.4535	0.2543	0.2740	0.1409	[8]
SLIM [25, 30]	0.7162	0.4468	0.8679	0.5601	[8]
iALS [20]	0.7111	0.4383	0.8762	0.5590	[8]
NeuMF (MLP+GMF) [17]	0.7093	0.4349	0.8777	0.5576	[8]
Matrix Factorization	**0.7294**	**0.4523**	**0.8895**	**0.5794**	Fig. 2

Learning dot product with MLP is hard:

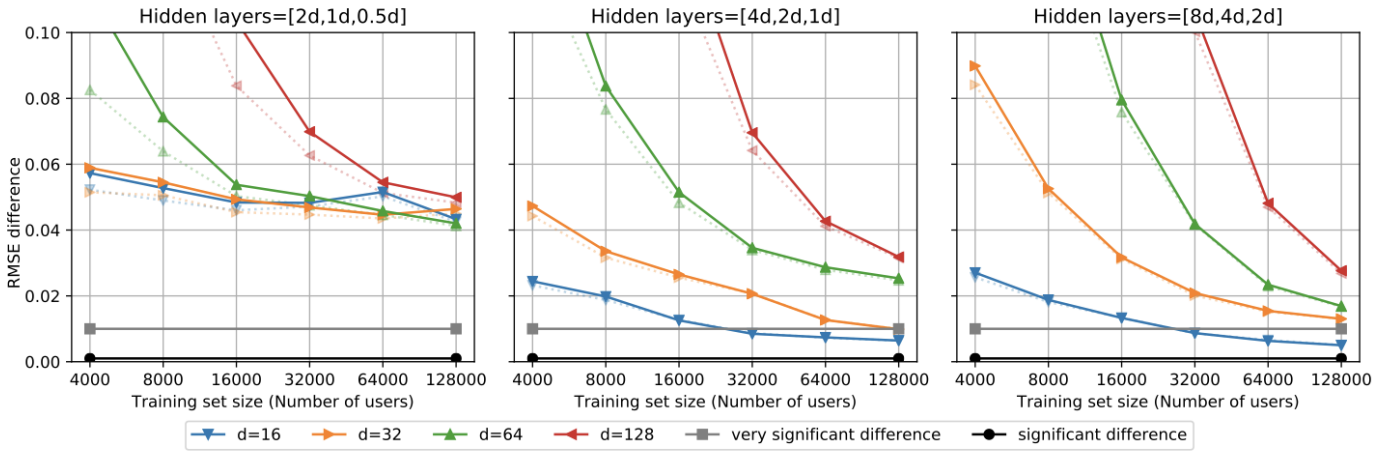


Figure 3: How well a MLP can learn a dot product over embeddings of dimension d. The ground truth is generated from a dot product of Gaussian embeddings plus Gaussian label noise. The graphs show the difference between the RMSE of the dot product and the RMSE of the learned similarity measure; the solid line measures the difference on the fresh set, the dotted on the test set. Noise and scale have been chosen such that 0.01 could indicate a very significant difference and 0.001 a significant difference.

Steffen Rendle, Walid Krichene, Li Zhang, and John Anderson. (RecSys '20). Neural Collaborative Filtering vs. Matrix Factorization Revisited

NeuroMF summary

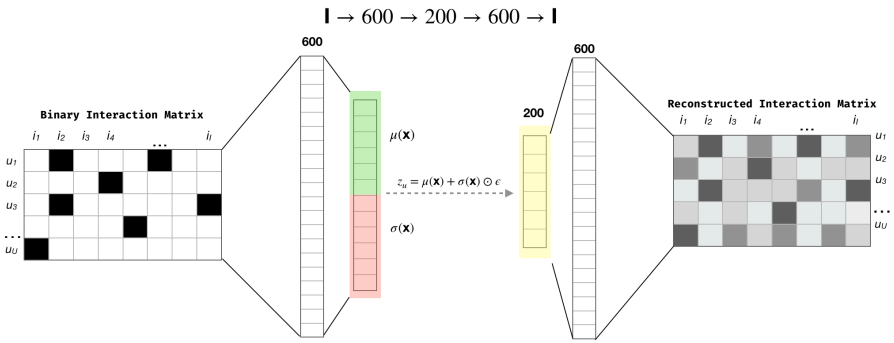
Pros

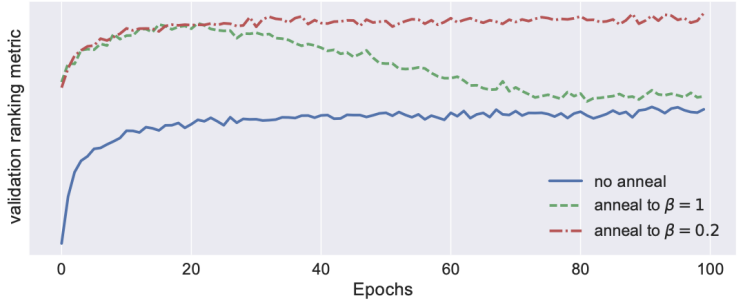
- Easy refitting on new interactions
- Used in production by Bytedance (tiktok owner)
  Z. Liu et al. (RecSys '22). Monolith: Real Time Recommendation System With Collisionless Embedding Table

Cons

- Has issues with outperforming matrix factorization
- Overall quality far from SOTA

MultVAE





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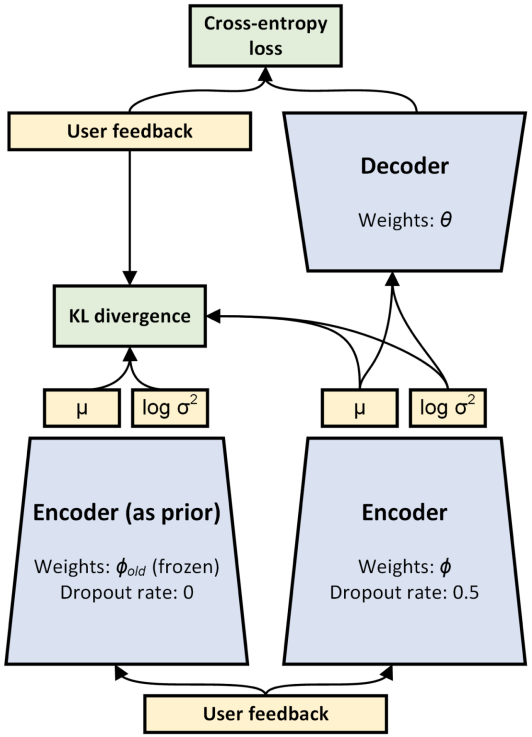
Idea: model user history with multinomial distribution

$$\mathcal{L}_\beta(\mathbf{x}_u; \theta, \phi) \equiv \mathbb{E}_{q_\phi(\mathbf{z}_u \mid \mathbf{x}_u)}[\log p_\theta(\mathbf{x}_u \mid \mathbf{z}_u)] - \beta \cdot \mathrm{KL}(q_\phi(\mathbf{z}_u \mid \mathbf{x}_u) \| p(\mathbf{z}_u)).$$

- The first term is the **reconstruction error** (multinomial likelihood).
- The second term is the **regularizer** (KL divergence from the prior distribution).

RecVAE



1. novel composite prior distribution
2. new approach to setting the hyperparameter $\beta = \beta(\mathbf{x}_u)$

	Recall@20	Recall@50	NDCG@100
MovieLens-20M Dataset			
WARP [38]	0.314	0.466	0.341
LambdaNet[6]	0.395	0.534	0.427
WMF [14]	0.360	0.498	0.386
SLIM [25]	0.370	0.495	0.401
CDAE [39]	0.391	0.523	0.418
Mult-DAE [22]	0.387	0.524	0.419
Mult-VAE [22]	0.395	0.537	0.426
RaCT [23]	0.403	0.543	0.434
EASE [33]	0.391	0.521	0.420
RecVAE (ours)	**0.414**±0.0027	**0.553**±0.0028	**0.442**±0.0021
Netflix Prize Dataset			
WARP [38]	0.270	0.365	0.306
LambdaNet[6]	0.352	0.441	0.386
WMF [14]	0.316	0.404	0.351
SLIM [25]	0.347	0.428	0.379
CDAE [39]	0.343	0.428	0.376
Mult-DAE [22]	0.344	0.438	0.380
Mult-VAE [22]	0.351	0.444	0.386
RaCT [23]	0.357	0.450	0.392
EASE [33]	**0.362**	0.445	0.393
RecVAE (ours)	0.361±0.0013	**0.452**±0.0013	**0.394**±0.0010

MultVAE summary

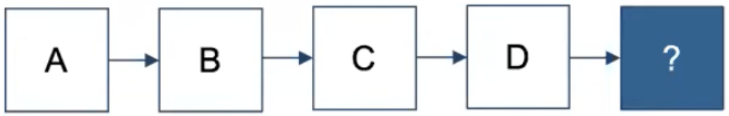
Pros

- Successfully reproduced in «*Are we really making much progress*»
- Closer to SOTA
- Predictions are from user's click history. No need to obtain latent factor for user.

Cons

- Handling cold items

Sequential recommendation

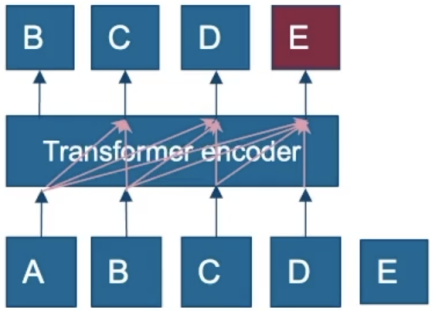


Task: predict next interaction in a set of user's interaction

Transformers for sequential recommendation

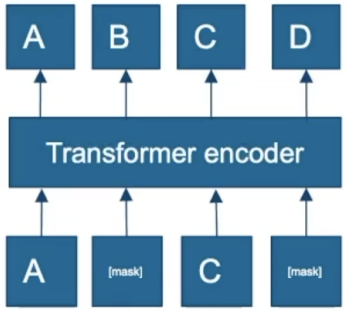
SASRec (Kang et al.):

- Predicts Input Sequence, shifted by 1 element
- Causal Self-Attention
- Use last prediction for recommendation



BERT4Rec (Sun et al.):

- Based on BERT language model
- Recover corrupted input for training
- Add [Mask] token to the end of the sequence for prediction



Sequential models comparison

Bert4Rec paper (CIKM'19):

Datasets	Metric	POP	BPR-MF	NCF	FPMC	GRU4Rec	GRU4Rec$^+$	Caser	SASRec	BERT4Rec	Improv.
Beauty	HR@1	0.0077	0.0415	0.0407	0.0435	0.0402	0.0551	0.0475	0.0906	**0.0953**	5.19%
	HR@5	0.0392	0.1209	0.1305	0.1387	0.1315	0.1781	0.1625	0.1934	**0.2207**	14.12%
	HR@10	0.0762	0.1992	0.2142	0.2401	0.2343	0.2654	0.2590	0.2653	**0.3025**	14.02%
	NDCG@5	0.0230	0.0814	0.0855	0.0902	0.0812	0.1172	0.1050	0.1436	**0.1599**	11.35%
	NDCG@10	0.0349	0.1064	0.1124	0.1211	0.1074	0.1453	0.1360	0.1633	**0.1862**	14.02%
	MRR	0.0437	0.1006	0.1043	0.1056	0.1023	0.1299	0.1205	0.1536	**0.1701**	10.74%
Steam	HR@1	0.0159	0.0314	0.0246	0.0358	0.0574	0.0812	0.0495	0.0885	**0.0957**	8.14%
	HR@5	0.0805	0.1177	0.1203	0.1517	0.2171	0.2391	0.1766	0.2559	**0.2710**	5.90%
	HR@10	0.1389	0.1993	0.2169	0.2551	0.3313	0.3594	0.2870	0.3783	**0.4013**	6.08%
	NDCG@5	0.0477	0.0744	0.0717	0.0945	0.1370	0.1613	0.1131	0.1727	**0.1842**	6.66%
	NDCG@10	0.0665	0.1005	0.1026	0.1283	0.1802	0.2053	0.1484	0.2147	**0.2261**	5.31%
	MRR	0.0669	0.0942	0.0932	0.1139	0.1420	0.1757	0.1305	0.1874	**0.1949**	4.00%

A.Petrov. A Systematic Review and Replicability Study of BERT4Rec for Sequential Recommendation (RecSys'22):

Dataset	Number of papers	BERT4Rec wins	SASRec wins wins	Ties
Amazon Beauty*	19	12 (63%)	5 (26%)	2 (11%)
ML-1M*	18	13 (72%)	3 (17%)	2 (11%)
Yelp	10	6 (60%)	4 (40%)	0 (0%)
Steam*	8	7 (88%)	1 (12%)	0 (0%)
ML-20M*	8	7 (88%)	0 (0%)	1 (12%)
Total (including all other datasets)	134	86 (64%)	32 (23%)	16 (12%)

IdGenRec (SIGIR'25)

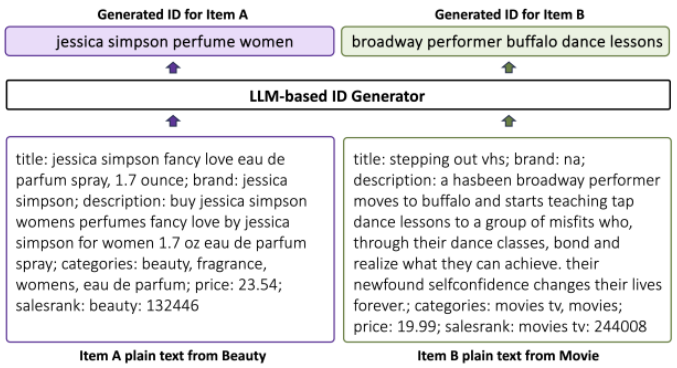


Figure 1: The ID generator takes plain text from each item's meta textual information and generates abstractive textual IDs for the item's representation.

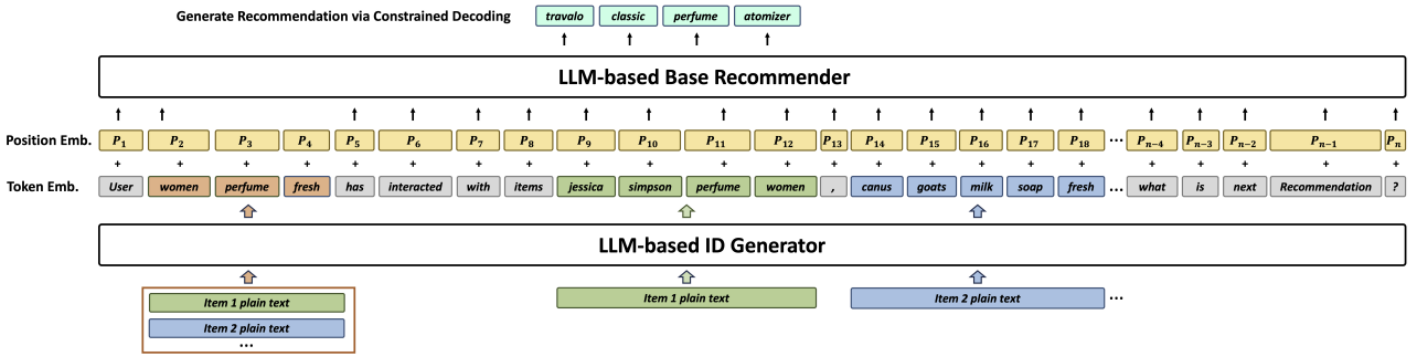


Figure 2: A real example showing the generative recommendation workflow. The ID generator generates item IDs for items from the user history by taking their plain text. Then, the generated IDs are interpolated into the template. Addtionally, the user ID is generated by using all items' text in the user's history, showing a "high-level profile" of the user's preference. The position embeddings are subsequently combined with token embeddings to capture the sequence of interactions. Finally, the base recommender generates the ID of the recommended item based on constrained decoding.

image source

Literature

1. *M. Dacrema, P. Cremonesi, D. Jannach.* (RecSys '19). Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches.
2. *X. He et al.* (WWW 2017). Neural Collaborative Filtering
3. *D. Liang et al.* (2020). Variational Autoencoders for Collaborative Filtering WWW '18.
4. *W. Kang, J. McAuley* (ICDM'19). BERT4Rec: Self-Attentive Sequential Recommendation
5. *F. Sun et al.* (CIKM'19). BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer
6. *A. Petrov, C. Macdonald* (RecSys '22). A Systematic Review and Replicability Study of BERT4Rec for Sequential Recommendation
7. *Tan J. et al.* (SIGIR'25). Idgenrec: Llm-recsys alignment with textual id learning